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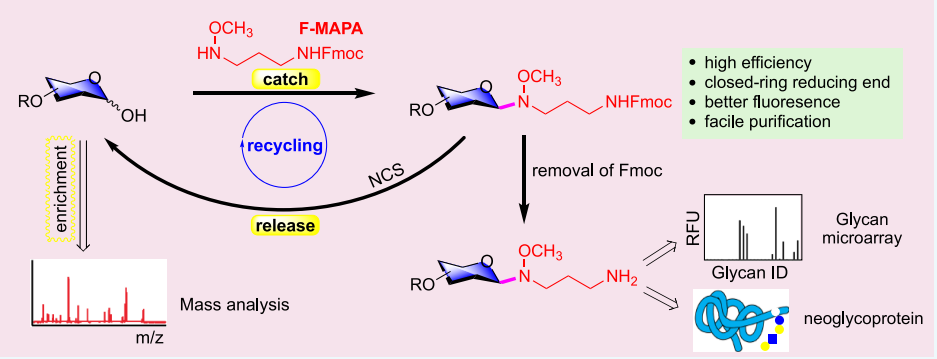
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Novel Reversible Fluorescent Glycan Linker for Functional Glycomics

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Supporting Information



ABSTRACT: To aid in generating complex and diverse natural glycan libraries for functional glycomics, more efficient and reliable methods are needed to derivatize glycans. Here we present our development of a reversible, cleavable bifunctional linker 3-(methoxyamino)propylamine (MAPA). As the fluorenylmethyloxycarbonate (Fmoc) version (F-MAPA), it is highly fluorescent and efficiently derivatizes free reducing glycans to generate closed-ring derivatives that preserve the structural integrity of glycans. A library of glycans were derivatized and used to generate a covalent glycan microarray using *N*-hydroxysuccinimide derivatization. The array was successfully interrogated by a variety of lectins and antibodies, demonstrating the importance of closed-ring chemistry. The glycan derivatization was also performed at large scale using milligram quantities of glycans and excess F-MAPA, and the reaction system was successfully recycled up to five times, without an apparent decrease in conjugation efficiency. The MAPA-glycan is also easy to link to protein to generate neoglycoproteins with equivalent glycan densities. Importantly, the MAPA linker can be reversibly cleaved to regenerate free reducing glycans for detailed structural analysis (catch-and-release), often critical for functional studies of undefined glycans from natural sources. The high conjugation efficiency, bright fluorescence, and reversible cleavage of the linker enable access to natural glycans for functional glycomics.

Complex carbohydrates (glycans) are essential constituents of all living organisms, occurring as both simple and complex structures in glycoproteins, proteoglycans, glycolipids, and as free glycans. Beyond their roles in energy storage and structural support, glycans via their interactions with glycan-binding proteins (GBPs) are important in numerous physiological and pathological processes, such as platelet clearance, cellular adhesion and migration, innate immune responses, fertilization, embryogenesis, pathogen infection, inflammation, and the development of autoimmune diseases and cancer.[1] Therefore, more and more therapeutic agents and diagnostic tools targeting on glycan−GBP interactions are under development.[2−7] Despite its well-recognized importance, glycomics, the structures and functions of glycans in biological systems, has lagged far behind genomics, transcriptomics, and proteomics. This lag is mainly due to the unique structural complexity of glycans, the nontemplate driven synthesis of glycans, and the indirect regulation of glycan synthesis by genes, which create unique technical challenges for glycan sequencing and synthesis, thereby hindering access to complex glycans for functional studies.

The ability to derivatize glycans to various supports, as in glycan microarray technologies and microbead presentations, has provided key insights into glycans recognition and studies to explore roles of glycans in cell adhesion and signaling, as well as recognition of glycans by viruses, antibodies, and various GBPs. These technologies require small quantities of glycans and permit rapid analysis of binding to hundreds of test glycans in relatively simple high-throughput assays formats.[8−12] The limitation of such technologies is the lack of diversity of synthetic glycans, which are limited because of the difficulties in glycan synthesis, as well as the questionable relevance of such limited repertoires of glycans to complex biological processes. One approach to circumvent these limitations is to acquire glycans from natural sources (e.g. organs, tissues, cells,

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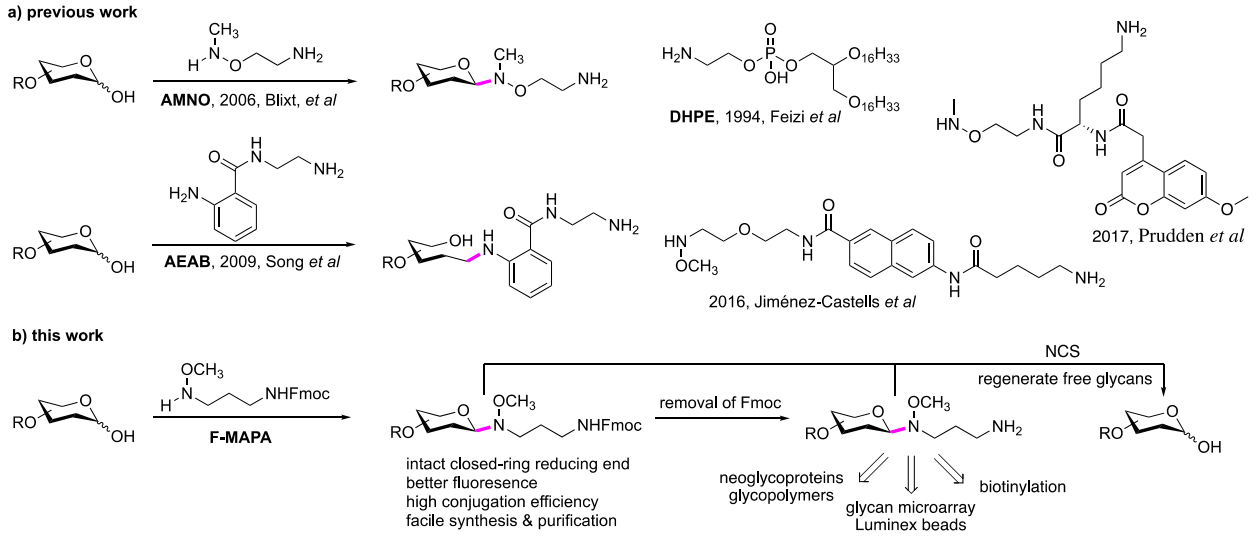


Figure 1. Rationale for linker design.

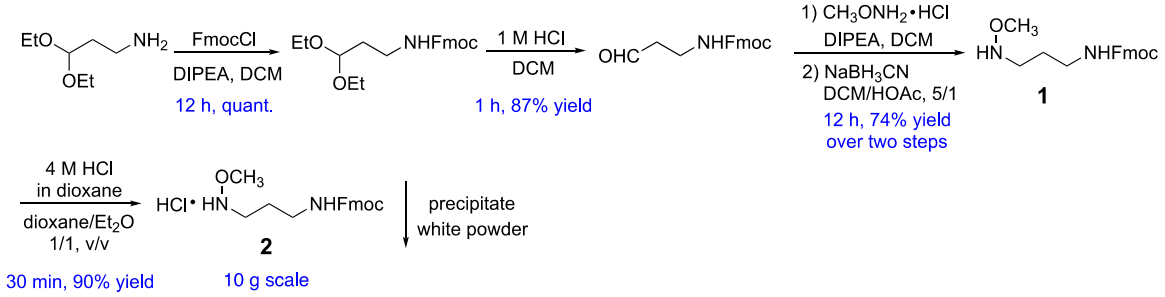


Figure 2. Synthesis of F-MAPA linker.

bacteria, etc.), which contain large glycomes of biological context and perhaps more relevant biological activities. However, while the isolation and purification of glycans from natural sources is a desirable strategy, there are many difficulties largely due to the lack of reversible and facile tagging methods for free, reducing glycans prepared from such sources.

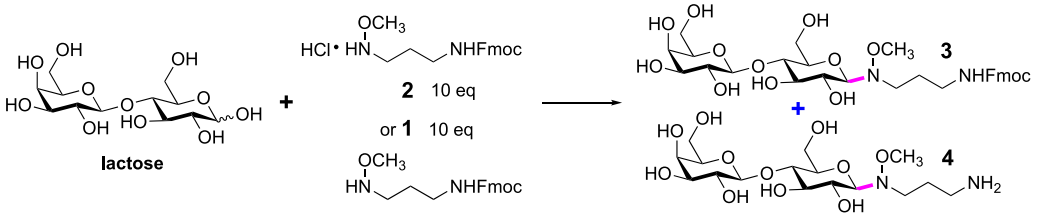
To utilize glycans from natural sources, glycans need to be derivatized with functional tags to enable separation and purification by multidimensional chromatography, detection, and quantification of purified glycans or subfractions, as well as subsequent immobilization on array surface and other applications.[13−17] Much effort has been devoted toward the development of linkers (or tags) for glycan derivatization.[18−22] Based on the chemistry of derivatization, these linkers can be classified into two categories: reductive amination and *N*-alkyl oxime ligation. Three representative linkers of these two categories are 1,2-dihexadecanoyl-*sn*-glycero-3-phosphoethanolamine (DHPE), 2-amino-N-(2-amino-ethyl)-benzamide (AEAB), and 2-[(methylamino)oxy]ethylamine (AMNO) (Figure 1a). DHPE and AEAB react with reducing glycans through reductive amination and open sugar ring of reducing end monosaccharide residue,[14,23] which destroys the reducing end integrity of glycans, which may affect their binding and the immunogenicity of glycans,[24] especially for small glycan epitopes. AMNO reacts with reducing glycans through *N*-alkyl oxime mediated selective ligation and forms a closed-ring reducing end, which preserves the integrity of glycans.[25] Nevertheless, AMNO lacks a fluorophore, making it difficult to detect, separate and quantify. Thus, the development of a fluorescent linker, which can derivatize glycans efficiently and preserve a closed-ring reducing end, is highly desired.

Recently, Jiménez-Castells et al. and Prudden et al. developed fluorescent *N*-alkyl oxime linkers which can form a closed-ring reducing end (Figure 1a).[26,27] However, these two linkers required complex synthesis and gave moderate conjugation yields. Moreover, long reaction time under acidic conditions (72 h, pH 4.6) rendered them incompatible with labile modifications (e.g., sulfation) and sialic acid residues typically found in glycans, thus substantially limiting their utility, especially for derivatization of natural glycans.

Here, we present a multifunctional fluorescent linker 3-(methoxyamino)propylamine (MAPA) with an Fmoc fluorescent addition, F-MAPA, for glycan derivatization, which can efficiently conjugate with glycans (synthetic and natural glycans) under mild conditions and form a closed-ring reducing end (Figure 1b). After conjugation with F-MAPA, the resulting Glycan-F-MAPA conjugates (GFMAPAs) can be purified by solid phase extraction (SPE), and Fmoc in the linker can be easily removed to generate an active alkyl amine enabling further manipulation and application of glycans for multiple purposes.[28−30] The linker can also be easily cleaved under mild conditions to regenerate free reducing glycans for detailed structure analysis by mass spectrometry (MS). The efficiency and reversibility of glycan derivatization as well as the enhanced detectability afforded by F-MAPA will promote novel studies of minor and major glycans in natural sources and their functional recognition by proteins.

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Table 1. Glycan and Linker Conjugation Condition Optimization



entry	linker	solvent	base	temp.	time	conversion[a]	yield[b] 3	yield[b] 4
1	**2**	MeOH/NaOAc buffer (pH 4.5), 1/1, v/v		40 °C	48 h	40%	33%	0%
2	**2**	DMSO/NaOAc buffer (pH 4.5), 1/1, v/v		40 °C	48 h	50%	24%	21%
3	**2**	DMSO		40 °C	48 h	< 5%	ND	ND
4	**2**	DMSO/HOAc 7/3, v/v		40 °C	48 h	40%	31%	0%
5	**1**	DMSO		40 °C	48 h	80%	53%	17%
6	**2**	DMSO	DIPEA, 3 eq	40 °C	48 h	75%	11%	52%
7	**2**	DMSO	NaOAc, 3 eq	40 °C	12 h	100%	0%	91%
8	**1**	DMSO/HOAc 7/3, v/v		40 °C	36 h	> 95%	89%	0%
9	**1**	DMSO/HOAc 7/3, v/v	DIPEA 10 eq	40 °C	36 h	> 95%	35%	50%
10	**1**	DMSO/HOAc 7/3, v/v	NaOAc 10 eq	40 °C	36 h	> 95%	52%	31%
11	**2**	DMSO/HOAc 7/3, v/v	DIPEA 10 eq	40 °C	36 h	> 95%	87%	0%
12	**2**	DMSO/HOAc 7/3, v/v	NaOAc 10 eq	40 °C	36 h	> 95%	88%	0%
13	**1**	DMSO/HOAc 7/3, v/v		65 °C	2 h	100%	90%	0%
14	**2**	DMSO/HOAc 7/3, v/v	DIPEA 10 eq	65 °C	2 h	100%	88%	0%
15	**2**	DMSO/HOAc 7/3, v/v	NaOAc 10 eq	65 °C	2 h	100%	91%	0%

[a]Based on lactose (5 mg). [b]Isolated yield; ND: not determined; eq: equivalent.

RESULTS

Synthesis of MAPA. We considered that *N*-alkyl oxime can stereoselectively couple with the hemiacetal of glycan reducing end to preserve the cyclic nature,[31] thus rendering development of fluorescent *N*-alkyl oxime linker highly attractive. To achieve high conjugation efficiency, the linker should be relatively small. In addition, its synthesis should be facile, efficient, scalable, and cost-effective to enable general accessibility to research laboratories and potential industrial application. In these regards, linker F-MAPA was designed, which contains an *N*-alkyl oxime active motif and an Fmoc-protected alkyl amine. Fmoc serves as a transient fluorophore for monitoring and quantification, as well as a hydrophobic tag for glycan enrichment, separation, and purification. As shown in Figure 2, F-MAPA (**1**) and its hydrochloride salt **2** were facilely and efficiently synthesized in 10 g scale.

Glycan and Linker Conjugation Condition Optimization. The conjugation of *N*-alkyl oxime with reducing glycans was generally conducted in pH 4.5 aqueous buffer. Nevertheless, due to poor solubility of **1** and **2** in water, combinations of aqueous buffer with methanol and dimethyl sulfoxide (DMSO), DMSO, DMSO/HOAc were tested as solvent for the conjugation of lactose and **2** (Table 1, entries 1−4). When MeOH/NaOAc buffer and DMSO/HOAc were used as solvent, 40% conversions were given and conjugation product **3** was provided exclusively with 33% and 31% yields, respectively (entries 1 and 4). DMSO/NaOAc buffer gave higher conversion and yield, but a mixture of **3** and **4** were generated (entry 2). The low conversion in DMSO (entry 3) might be attributed to its low polarity, which suppressed the dissociation of protonated *N*-alkyl oxime, thereby blocking ligation of *N*-alkyl oxime to reducing end hemiacetal of glycans. Thus, unprotonated **1** or **2** with base additive were screened (entries 5−7). Indeed, conversions and yields were substantially increased in all three scenarios. Nevertheless, **1** or **2** with *N*,*N*-diisopropylethylamine (DIPEA) gave a mixture of **3** and **4** (entries 5 and 6), whereas **2** with NaOAc provided **4** exclusively (entry 7). According to the results above, DMSO/HOAc provides **3** exclusively, and base increases conversion and yield. Therefore, DMSO/HOAc was chosen as a solvent,

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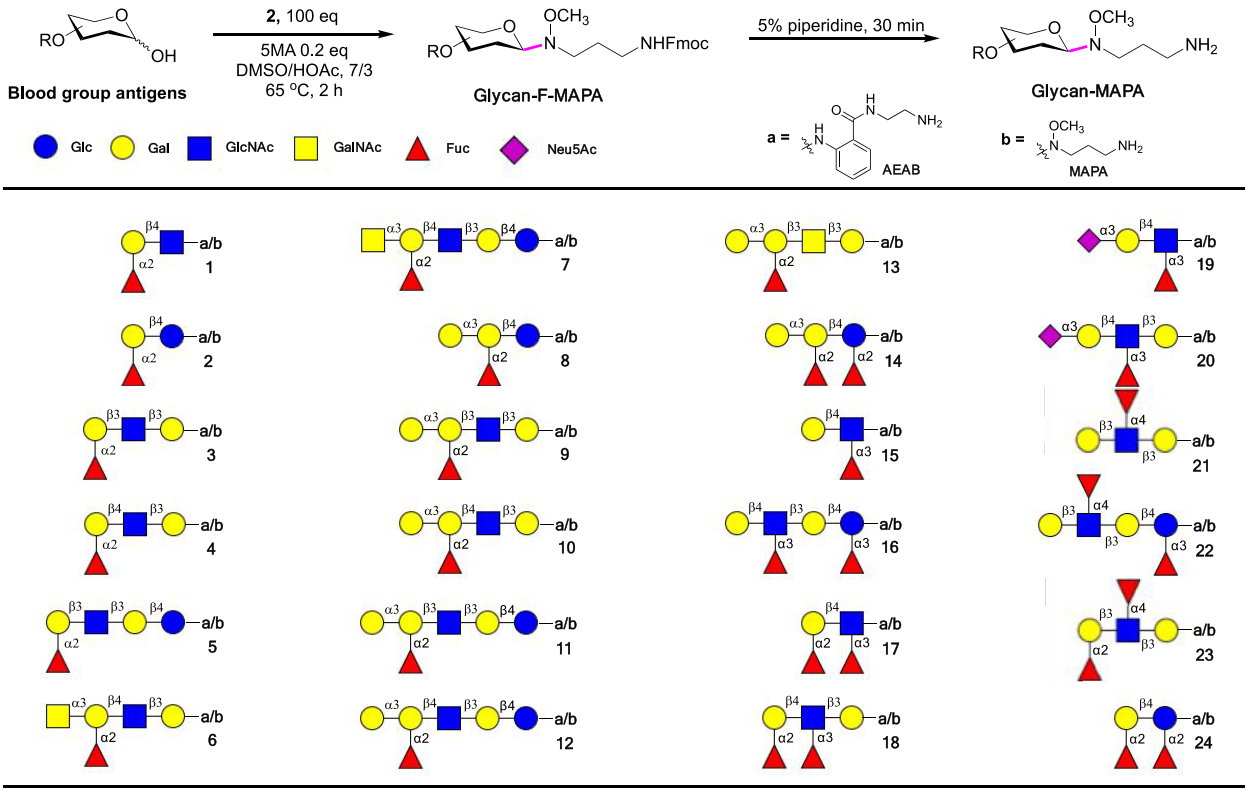


Figure 3. Derivatization of blood group glycans with F-MAPA.

and **1** or **2** with base additive was used (entries 8−12). When **1** was used as linker, **3** was provided exclusively with 89% yield (entry 8), whereas addition of base led to formation of a mixture of **3** and **4** (entries 9 and 10). For **2**, addition of base gave higher yields, and **3** was provided exclusively (entries 11 and 12 vs entry 4). When conjugation temperature was increased to 65 °C, reaction time was substantially reduced to 2 h, and **3** was provided with high yields (entries 13−15). Nevertheless, in consideration of conjugation efficiency, convenient linker handling and storage, the combination of **2** (white powder) and NaOAc were chosen. Conjugation conditions were set to DMSO/HOAc as solvent and heating at 65 °C for 2 h (entry 15), which had been demonstrated to be compatible with natural glycans in previous works.[14,32,33]

Derivatization of Reducing Glycans. To demonstrate the utility of F-MAPA linker, 24 commercially available blood group glycans were derivatized with **2** in microgram scale. 5-Methoxyanthranilic acid (5MA) was reported to promote ligation of *N*-alkyl oxime and free reducing glycans.[34,35] To ensure complete conversion of glycans, 100 equiv of **2** was used, and 0.2 equiv of 5MA was added. The reaction mixture was shaken at 65 °C for 2 h, and then 10 vols of ethyl acetate was added to precipitate GFMAPAs. The precipitates were collected and purified by C-18 SPE to provide GFMAPAs with high yields (Figure 3). The 24 blood group glycans were also derivatized with AEAB for comparison and further evaluation of F-MAPA linker in microarray analysis.

Microarray Analysis of F-MAPA- and AEAB-labeled Glycans. Fmoc was found to be a good hydrophobic tag for reverse-phase purification of GFMAPAs. We tried to print GFMAPAs directly on nitrocellulose and polystyrene slides, but unfortunately, GFMAPAs do not attach to either slide, although a wide range of concentrations were tested (from 50 μM up to 1 mM). Thus, we removed the Fmoc by piperidine to give MAPA-labeled glycans in quantitative yield, which provided active alkyl amine and allowed printing on NHS-activated slides. MAPA- and AEAB-labeled glycans were printed on NHS-activated slides at the same concentration (50 μM). Probe **1a**-**24a** were AEAB-labeled blood group glycans with open-ring reducing ends. Probes **1b**−**24b** were MAPA-labeled blood group glycans with closed-ring reducing ends. Probes **1**−**5** were H antigens, **6**−**7** were A antigens, **8**−**14** were B antigens, and **15**−**24** were Lewis antigens and analogues. Probe **25a** is biotinylated hydrazide, and probe **25b** is sodium phosphate. The quality of the array was validated by specificity well-defined plant lectins, and then the array was interrogated by antiblood group antibodies.

Aleuria aurantia lectin (AAL) and *Ulex europaeus agglutinin* I (UEA-I) are fucose binding lectins. AAL recognizes terminal fucose residues and core fucose residues,[36] whereas UEA-I is specific for α1−2-linked fucose of terminal Fucα1−2Galβ1−4GlcNAc/Glc determinants.[37] As expected, both AEAB- and MAPA-labeled blood group H glycans (**1**−**5**) and Lewis antigens (**15**−**24**) exhibited strong binding with AAL (Figure 4a), whereas blood group A glycans (**6** and **7**) and blood group B glycans (**9**−**13**) generally did not bind AAL. Interestingly, although both AEAB- and MAPA-labeled **8** and **14** bound to AAL, the binding of AEAB-labeled **8b** and **14b** with AAL was much stronger, which might result from increased accessibility of fucose residues by AAL due to open-ring reducing end. In agreement with previous reports, only H2 and H5 glycans (**1**, **2** and **4**), Le^y (**17** and **18**), and 2,2′-FL (**24**) were recognized by UEA-I (Figure 4b), and the binding of **17a** is much weaker than that of **17b**. *Griffonia simplicifolia* I−B4 isolectin (GSL-I B4) and *Dolichos biflorus* lectin (DBA) recognize terminal Galα-[38] and GalNAcα1−3GalNAc/Gal,[39,40] respectively.

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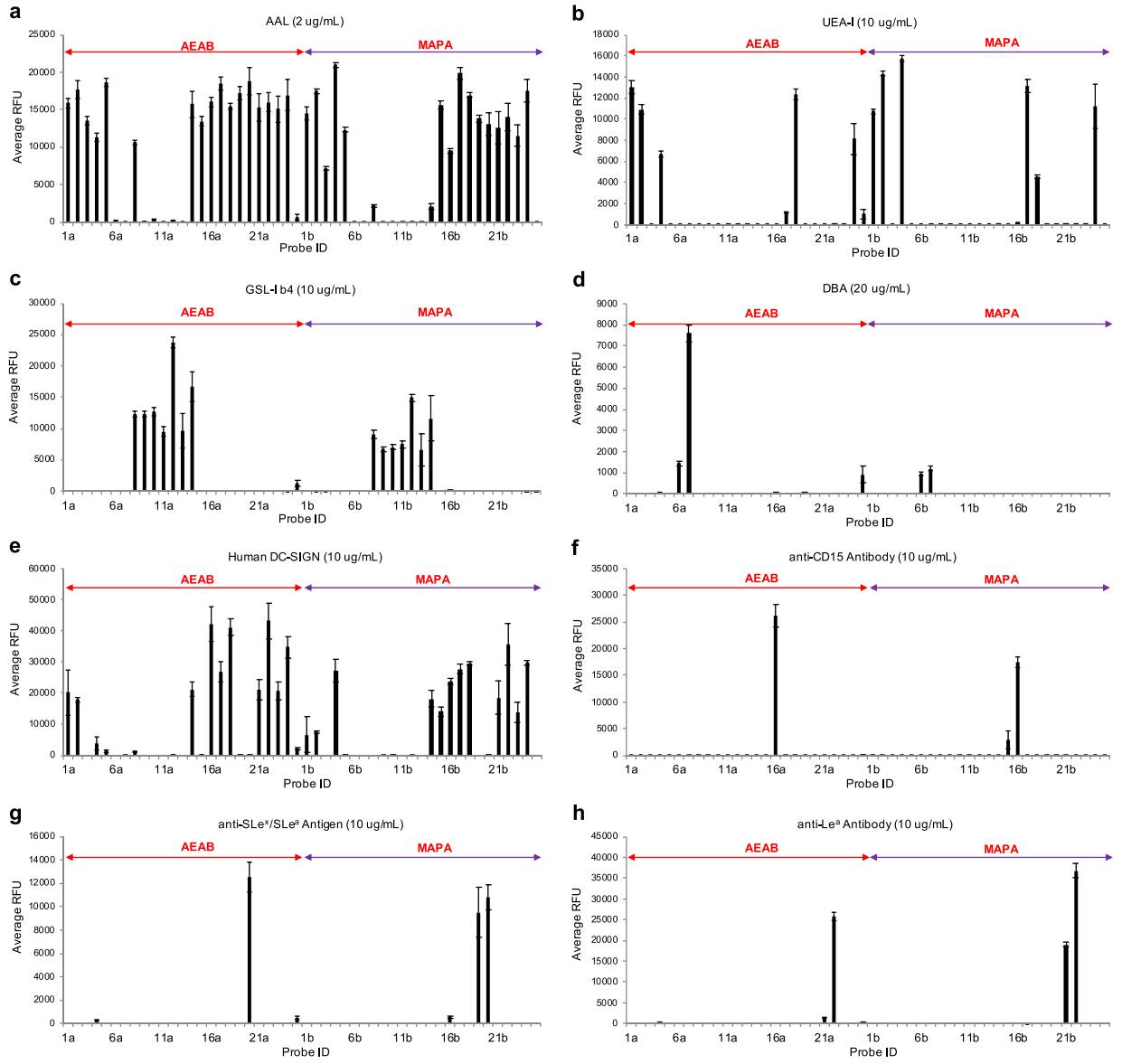


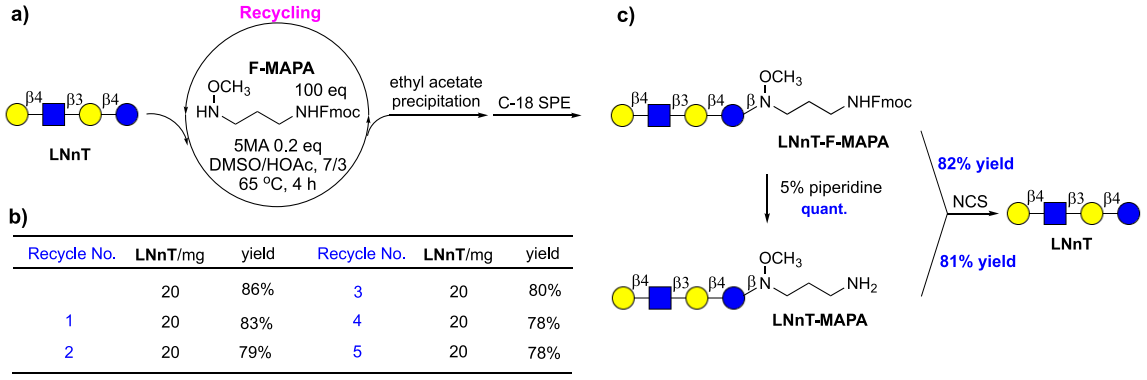
Figure 4. Recognition of blood group antigens by plant lectins and antiblood group antibodies.

Therefore, only blood group B glycans (**8**−**14**) were bound by GSL-I B4 (Figure 4c), and blood group A glycans (**6** and **7**) were bound by DBA (Figure 4d). Altogether, the lectin studies demonstrated that all probes were well printed on the array and that overall the lectin binding patterns of AEAB- and MAPA-labeled glycans were similar or almost the same.

To further investigate the influence of the linker on glycan recognition by GPBs, the glycan microarray was interrogated by human dendritic cell-specific intercellular adhesion molecule-3-grabbing nonintegrin (DC-SIGN), and monoclonal anti-CD15, anti-SLe^x/SLe^a, anti-Le^a antibodies. DC-SIGN bound to H2, H5 (**1**, **2**, **4**) and Lewis glycans (**15**−**18** and **21**−**23**), which corroborated previous studies on the glycan specificity of DC-SIGN.[41−43] Notably, glycan **14** and **24** exhibited strong binding, whereas glycan **8** were not bound by DC-SIGN (Figure 4e), which suggested that 2,2′-FL was a new binding motif of DC-SIGN. Moreover, DC-SIGN bound to MAPA-labeled Le^x trisaccharide **15b** but not AEAB-labeled Le^x trisaccharide **15a**, indicating the importance of preserving intact closed-ring reducing end for glycan recognition. Similarly, anti-CD15 antibody and anti-SLe^x/SLe^a antibody bound to MAPA-labeled **15b** and **19b**, respectively, but not AEAB-labeled **15a** or **19a** (Figure 4f,g). In addition, MAPA-labeled **21b** exhibited strong binding, but AEAB-labeled **21a** was weakly bound by the anti-Le^a antibody (Figure 4h). These preferences toward MAPA-labeled glycans further highlight the importance of preserving intact closed-ring reducing ends for glycan recognition by GBPs.

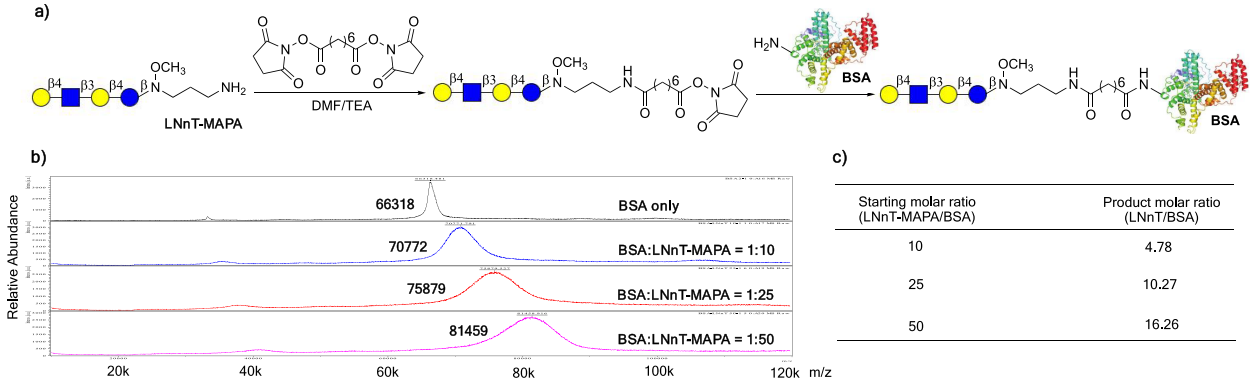
Reaction System Recycling and Cleavage of Linker to Regenerate Free Reducing Glycans. Given the high conjugation efficiency and facile purification procedure, we envisioned that F-MAPA could be applied in large-scale glycan derivatization, and the reaction system could be recycled. As a demonstration, LNnT was derivatized with F-MAPA in preparative scale. LNnT (20 mg) was dissolved in DMSO/HOAc, and then 100 equiv of F-MAPA and 0.2 equiv of 5MA were added. After the solution was stirred at 65 °C for 4 h, LNnT was completely consumed. Then, 10 vols of ethyl acetate was added to precipitate the LNnT-F-MAPA conjugate. The precipitate was collected by centrifugation

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Recycle No.	LNnT/mg	yield	Recycle No.	LNnT/mg	yield
	20	86%	3	20	80%
1	20	83%	4	20	78%
2	20	79%	5	20	78%

Figure 5. (a) Recycling of reaction system. (b) Conjugation efficiency of F-MAPA in recycling. (c) cleavage of linker to regenerate free reducing glycans.



c)

Starting molar ratio (LNnT-MAPA/BSA)	Product molar ratio (LNnT/BSA)
10	4.78
25	10.27
50	16.26

Figure 6. Synthesis of neoglycoprotein. (a) Synthetic scheme. (b) Mass spectra of LNnT-BSA conjugates. (c) Average number of glycan on BSA.

and purified by C-18 SPE to give LNnT-F-MAPA with 86% yield. The supernatant was concentrated to remove ethyl acetate, and 20 mg LNnT was added once again (Figure 5a). The mixture was stirred at 65 °C for another 4 h, and LNnT-F-MAPA was generated with 83% yield. The reaction system was recycled for 5 times, and there was no apparent deterioration in the conjugation efficiency, with the yield of LNnT-F-MAPA ranging from 86% in the first cycle to 78% in the sixth cycle (Figure 5b). In addition, after 6 cycles, linker F-MAPA was also recovered with 80% yield. The successful recycling of reaction system and recovery of linker render linker F-MAPA hold great potential for industrial-scale application.

F-MAPA/MAPA tag can increase MS sensitivity compared with unmodified glycans; nevertheless, it also complicates permethylation of glycans and limits more detailed structural analysis by MS, which is critical for the functional study of undefined glycans from natural sources. Anomeric *N*-alkyl oxime can be hydrolyzed under acidic conditions;[44,45] however, the challenge was to establish conditions to cleave the *N*-alkyl oxime linker without affecting the labile fucose and sialic acid residues. *N*-Chlorosuccinimide (NCS) was reported to cleave anomeric *N*-alkyl oxime with high yield; nevertheless, only protected mono- or disaccharides were tested.[46] Therefore, NCS was explored for the cleavage of F-MAPA linker. LNnT-F-MAPA and LNnT-MAPA were treated with NCS in water. Importantly, both F-MAPA and MAPA were cleanly cleaved by NCS in 2 h, and LNnT (confirmed by MS and NMR) was regenerated with 82% and 81% yield, respectively (Figure 5c and Figure S1). To further demonstrate the robustness of NCS mediated cleavage of *N*-alkyl oxime linker, fucosylated and sialylated **20b** was incubated with NCS in water for 2 h. The F-MAPA linker was selectively removed, and reducing glycan **20** was provided in 80% yield without affecting labile sialic acid and fucose residues (Figure S2). The ability to add and remove F-MAPA in a facile manner brings versatility to F-MAPA-derivatized glycans, facilitating both glycomics studies and functional glycan recognition.

Synthesis of Neoglycoprotein. After removal of Fmoc, an active alkyl amine is generated, which enables further manipulation and application of MAPA-labeled glycans. As an example, LNnT-MAPA was applied in the synthesis of neoglycoprotein. LNnT-MAPA was first conjugated to NHS-activated suberic acid spacer, and then the LNnT-suberate conjugate was coupled to alkyl amine side chain of lysine residues on bovine serum albumin (BSA) to form neoglycoprotein (Figure 6a). Since the conjugation of LNnT-MAPA and NHS-activated suberic acid was quantitative, LNnT-suberate was conjugated with BSA at different molar ratios to further evaluate the conjugation of LNnT-MAPA with proteins. As expected, the average number of glycan conjugated on per BSA is dependent on the ratio of LNnT-MAPA to BSA. At 10:1 ratio, ~4−5 LNnT/BSA were conjugated; at 25:1 ratio, ~10−11 LNnT/BSA were conjugated; at 50:1 ratio, ~16−17 LNnT/BSA were conjugated (Figures 6b,c and S3).

Derivatization of Natural Glycans. To further demonstrate the utility of F-MAPA linker, the same amount of crude egg yolk *N*-glycans (released by ORNG method,[47] 1.5 mg) was conjugated with F-MAPA versus AEAB. F-MAPA- and AEAB-labeled egg yolk *N*-glycans were quantified according to their fluorescence, and F-MAPA gave slightly higher yield (0.201 μmol vs 0.193 μmol, Figure 7a). F-MAPA- and AEAB-labeled

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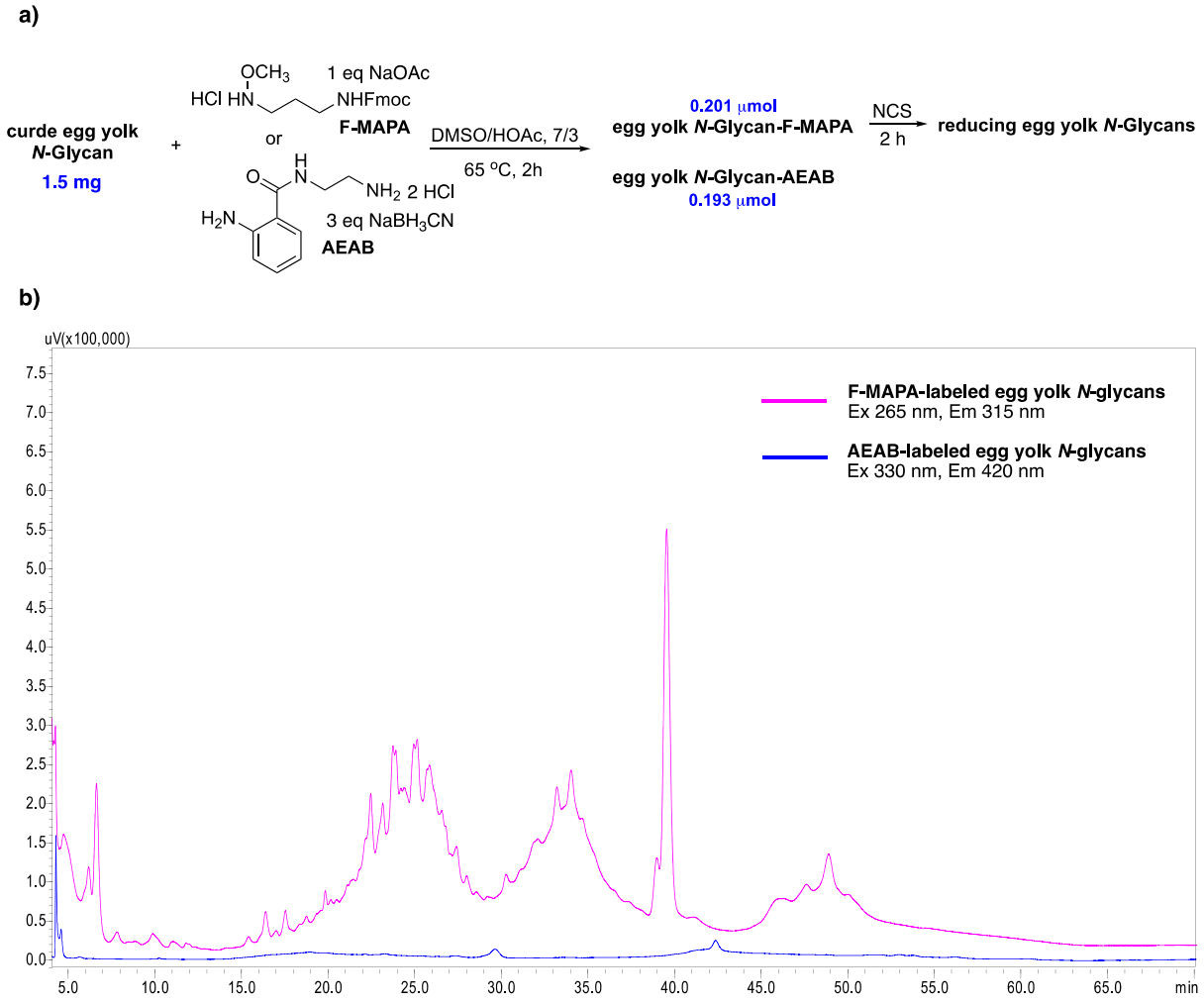


Figure 7. (a) Derivatization of crude egg yolk *N*-glycans with F-MAPA and AEAB. (b) HPLC profiles of F-MAPA- and AEAB-labeled egg yolk *N*-glycans.

N-glycans were normalized to the same concentration and injected into HPLC (NH_2 HILIC column). As shown in Figure 7b, the fluorescence intensity of F-MAPA was substantially higher than that of AEAB. In addition, according to the fluorescent standard curves of F-MAPA -, AEAB-, and anthranilamide (2-AB)-labeled LNnT, F-MAPA was 18-fold brighter than AEAB and 16-fold brighter than 2-AB (Figure S4), which could enable more sensitive detection and quantification during isolation, purification, and LC/MS analysis of F-MAPA-labeled natural glycans, thereby facilitating the development of natural glycan microarray. Then, F-MAPA-labeled egg yolk *N*-glycans were treated by NCS. F-MAPA was selectively removed, and reducing egg yolk *N*-glycans were regenerated. This catch-and-release chemoselectively isolated reducing egg yolk *N*-glycans from the crude mixture, which suggested that F-MAPA could potentially be used for the enrichment of reducing glycans in biological samples (urine, blood, human milk, etc.).

■ DISCUSSION

Our results demonstrate that we have successfully developed a highly efficient, fluorescent, and reversible tagging approach for reducing glycans that promotes their detection, separation, and analysis, as well as providing a linker for covalently coupling glycans to other molecules, including proteins and biotin. The fluorescent linker F-MAPA is easy to synthesize and its fluorescence is more sensitive than conventional fluorophores by many folds, such as 2-AB. The catch-and-release advantages of F-MAPA, as well as the reversibility of tagging, provides an unprecedented opportunity to detect and functionally assess minor quantities of glycans from natural sources. The availability of glycans from natural sources using F-MAPA as a linker will enable both glycan analytical and purification approaches and enlarge the availability of glycans to explore for functional recognition.

Despite the well-recognized importance of glycans in numerous physiopathological processes and medicine, functional glycomics study has been impeded by limited access to complex glycans. Tremendous progress has been achieved in the synthesis of glycans,[48−52] and several defined glycan microarrays with large glycan libraries have been developed. Nevertheless, the synthetic approach still lags far behind the explosive need for more diverse and biologically relevant glycans for biomedical research. Synthesis of modified mammalian glycans (phosphorylation, sulfation, etc.), plant, and microbial glycans is still challenging because of a lack of general chemical glycosylation protocols[53,54] and limited types of glycosyltransferases available at the current stage.[55,56] More importantly, the synthetic approach is target-driven, but often the actual biological target is unknown. On the other hand,

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glycans from a natural source (tissue, bacteria, cells, organisms) contain the entire glycome of interest and are directly implicated in the biological issues probed.

The power of natural glycan microarrays has been highlighted in several reports,[32,33,57,58] and one of the key aspects for development of natural glycan microarrays is efficient and reliable method to derivatize glycans. The F-MAPA linker described here can efficiently derivatize glycans and facilitate glycan-F-MAPA conjugate separation/purification. Synthesis of the linker was facile, efficient, and cost-effective, with 10 g of F-MAPA being synthesized in one batch in a few days. Considering operational simplicity of the synthetic design, the synthesis can easily be reproduced and scaled up in other research laboratories. The linker will also be available to the research community via our center with cost for shipping and handling to ensure general accessibility.

To demonstrate the utility of F-MAPA, blood group glycans were derivatized with F-MAPA in microgram scale and printed on NHS-activated slides. The resulting glycan microarray was validated by specific well-defined plant lectins. The lectin binding patterns of AEAB- and MAPA-labeled glycans were similar or almost the same. Then, the array was interrogated by antiblood group antibodies. 2,2′-FL was identified as a new binding motif of human DC-SIGN. The binding preference toward MAPA labeled **15b**, **19b**, and **21b** by DC-SIGN and anti-Lewis antibodies highlighted the importance of preserving intact closed-ring reducing end for glycan recognition by GBPs.

Given the high conjugation efficiency and facile purification procedure of F-MAPA mediated glycan derivatization, LNnT was conjugated with F-MAPA in preparative scale, and the reaction system was recycled for 5 times. The yield of LNnT-F-MAPA conjugate ranged from 86% in the first cycle to 78% in the sixth cycle, without apparent deterioration in the conjugation efficiency. In addition, after 6 cycles, the linker F-MAPA was also recovered with 80% yield. After removal of Fmoc, active alkyl amine is generated, enabling further manipulation and application of MAPA-labeled glycans. LNnT-MAPA was coupled to BSA to synthesize neoglycoproteins. The average number of glycans conjugated on per BSA is dependent on the ratio of LNnT-MAPA to BSA. The successful recycling of the reaction system and synthesis of neoglycoproteins render F-MAPA hold great potential for industrial synthesis of neoglycoprotein therapeutics and vaccines.

To further investigate the scope of F-MAPA-mediated glycan derivatization, the same amount of crude egg yolk *N*-glycans (released by bleach) was conjugated with F-MAPA and AEAB separately. F-MAPA gave slightly higher yield (0.201 μmol vs 0.193 μmol) than AEAB, and the fluorescence of F-MAPA was 18-fold brighter than that of AEAB, which could enable efficient derivatization, more sensitive detection, and quantification of F-MAPA-labeled natural glycans, thereby facilitating the development of natural glycan microarray. Upon treatment of F-MAPA-labeled glycans with NCS, F-MAPA was removed, and reducing egg yolk *N*-glycans were regenerated, thereby enabling detailed glycan structural analysis by MS. This catch-and-release chemoselectively isolated reducing egg yolk *N*-glycans from the crude mixture, which suggested that F-MAPA could potentially be used for the enrichment of reducing glycans in biological samples (urine, blood, human milk, etc.).

CONCLUSIONS

Natural glycans from biological sources contain the whole glycome of interest and are more biologically relevant. Expanding the glycan microarray technology with natural glycans represents the most attractive strategy to access complex and diverse glycan libraries for functional glycomics. For this approach to be effective in practice, an efficient and reliable method to derivatize glycans is needed. We developed a multifunctional fluorescent linker F-MAPA, which can efficiently derivatize reducing glycans under mild conditions and preserve the structure integrity. F-MAPA can also be facilely cleaved under mild conditions to regenerate free reducing glycans for detailed structure analysis by MS, which is the key for functional glycomics of undefined natural glycan microarrays. This catch-and-release approach also allows selectively enrichment of free reducing glycans from biological samples, which has vast potential applications in the bioanalysis and diagnosis. The facile and scalable synthesis of F-MAPA, high conjugation efficiency, and operational simplicity of F-MAPA mediated glycan derivatization make the linker accessible to general research laboratories and allow derivatization of glycans by biologists, thereby enabling general accessibility of glycan microarrays and customized development of glycan microarrays by researchers themselves. Moreover, the successful recycling of reaction system and synthesis of neoglycoprotein render F-MAPA hold great potential for its industrial application in the synthesis of neoglycoprotein therapeutics and vaccines.

METHODS

Synthesis of F-MAPA. To a stirred solution of 3-aminopropionaldehyde diethylacetal (7.36 g, 50 mmol, 1 equiv) and *N,N*-diisopropylethylamine (DIPEA, 12.93 g, 100 mmol, 2 equiv) in dichloromethane (DCM, 250 mL), Fmoc-Cl (13.58 g, 52.5 mmol, 1.05 eq in 30 mL DCM) was added dropwise at 0 °C. After completion of addition, the mixture was allowed to warm to rt and stirred overnight. The reaction mixture was washed with 1 M HCl, dried over Na_2SO_4, and concentrated to provide Fmoc-3-amino propionaldehyde diethylacetal with quantitative yield, which was sufficiently pure and used directly.

Fmoc-3-amino propionaldehyde diethylacetal was dissolved in 150 mL DCM, and then 4 M HCl in dioxane (30 mL) was added at −78 °C and stirred for 1 h. The mixture was diluted with DCM (150 mL) and washed with aqueous Na_2CO_3 (1 M, 200 mL). After phase separation and extraction of the aqueous phase with DCM (100 mL), the combined organic phases were dried over Na_2SO_4, concentrated under reduced pressure and purified by flash chromatography (silica gel column, hexane/ethyl acetate, from 0 to 35%) to afford Fmoc-3-amino propionaldehyde as a white solid (12.85 g, 87% yield). ^{1}H NMR (400 MHz, $CDCl_3$): δ 9.80 (s, 1 H), 7.76 (d, J = 7.6 Hz, 2 H), 7.57 (d, J = 7.6 Hz, 2 H), 7.40 (m, 2 H), 7.31 (m, 2 H), 5.21 (s, 1 H), 4.39 (d, J = 6.8 Hz, 2 H), 4.20 (t, J = 6.8 Hz, 1 H), 3.49 (dd, J = 12.0, 6.0 Hz, 2 H), 2.72 (t, J = 6.0 Hz, 2 H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 201.1, 156.3, 143.8, 141.3, 127.6, 127.0, 125.0, 119.9, 66.6, 47.2 44.0, 34.4.

To a solution of Fmoc-3-amino propionaldehyde (12.85 g, 43.5 mmol, 1 equiv) and CH_3ONH_3Cl (5.45 g, 65.25 mmol, 1.5 equiv) in DCM (150 mL), DIPEA (16.83 g, 130.5 mmol, 3 equiv) was added. The mixture was stirred overnight at rt and then concentrated. The residue was dissolved in DCM/HOAc

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(4/1, v/v, 200 mL), and then $NaBH_3CN$ (4.11 g, 65.25 mmol, 1.5 equiv) was added at 0 °C in several portions and stirred for 1 h. The reaction mixture was concentrated and the residue was purified by flash chromatography (silica gel column, hexane/ethyl acetate, from 0 to 50%) to afford F-MAPA (**1**) as a colorless oil (10.51 g, 74% yield). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.76 (d, J = 7.6 Hz, 2 H), 7.59 (d, J = 7.6 Hz, 2 H), 7.40 (dd, J = 7.6, 7.6 Hz, 2 H), 7.31 (dd, J = 7.6, 7.6 Hz, 2 H), 5.20 (s, 1 H), 4.41 (d, J = 6.8 Hz, 2 H), 4.21 (t, J = 6.8 Hz, 1 H), 3.54 (s, 1 H), 3.30 (t, J = 6.4 Hz, 2 H), 2.97 (t, J = 6.4 Hz, 2 H), 1.73 (tt, J = 6.4, 6.4 Hz, 2 H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 156.5, 144.0, 141.3, 127.6, 127.0, 125.0, 119.9, 66.5, 61.9, 49.5, 47.3, 39.4, 27.5. ESI HRMS, calcd for $C_{19}H_{23}N_2O_3$ $[M + H]^+$ 327.1703, found 327.1711.

To a solution of F-MAPA (10.51 g, 32.19 mmol) in dioxane (50 mL), 4 M HCl in dioxane (10 mL) was added dropwise at 0 °C. Then, cold Et_2O was added until no more precipitate formed. The precipitate was collected and washed with cold Et_2O to afford F-MAPA hydrochloride salt **2** as a white powder (10.51 g, 90% yield). ^{1}H NMR (400 MHz, MeOD): δ 7.77 (d, J = 7.6 Hz, 2 H), 7.62 (d, J = 7.6 Hz, 2 H), 7.37 (dd, J = 7.6, 7.6 Hz, 2 H), 7.29 (dd, J = 7.6, 7.6 Hz, 2 H), 4.37 (d, J = 6.8 Hz, 2 H), 4.17 (t, J = 6.8 Hz, 1 H), 3.91 (s, 1 H), 3.28 (t, J = 7.6 Hz, 2 H), 2.97 (t, J = 6.4 Hz, 2 H), 1.89 (tt, J = 7.6, 6.4 Hz, 2 H); ^{13}C NMR (100 MHz, MeOD) δ 157.7, 143.8, 141.2, 127.4, 126.7, 124.7, 119.6, 66.3, 60.9, 47.1, 46.5, 37.2, 24.1.

Conjugation of Free Oligosaccharides with 2 and AEAB. 0.35 M **2**, 1 M NaOAc (with 2 mM 5MA), 0.35 M AEAB and 1 M $NaBH_3CN$ in DMSO/AcOH (7:3 v/v) were freshly prepared. To 100 μg of glycan, 100 equiv of **2**, 100 equiv of NaOAc, and 0.2 equiv of 5MA were added. The mixture was vortexed and heated at 65 °C on a shaker for 2 h. Then, 10 vols of ethyl acetate was added, vortexed, and frozen at −20 °C for 20 min to precipitate the glycan-F-MAPA conjugate. The precipitate was collected by centrifugation and dried in a Centra-vap evaporator. The dried sample was dissolved in 150 μL of water and loaded on a C18 SPE cartridge (50 mg). The cartridge was washed with 5 × 1 mL of water, and the glycan-F-MAPA conjugate was eluted by 4 × 0.6 mL 30% acetonitrile (ACN). Fractions were combined and lyophilized. Glycan-F-MAPA was quantified by fluorometer (Ex 265 nm, Em 315 nm).

After quantification, glycan-F-MAPA was treated with 5% piperidine in water (500 μL) for 30 min, and MALDI analysis showed complete removal of Fmoc. The mixture was washed with chloroform for 3 times, and the aqueous phase was loaded on a C18 SPE cartridge (50 mg). The cartridge was eluted with 4 × 0.6 mL of water. Fractions were collected and lyophilized to provide glycan-F-MAPA conjugate with quantitative yield.

To 100 μg glycan, 200 equiv of AEAB and 600 equiv $NaBH_3CN$ were added. The mixture was vortexed and heated at 65 °C on shaker for 2 h. Then, 10 vols of ACN was added, vortexed, and frozen at −20 °C for 20 min to precipitate glycan-AEAB conjugate. The precipitate was collected by centrifugation. The sample was dissolved in 150 μL of 85% ACN and loaded on a NH_2 SPE cartridge (100 mg). The cartridge was washed with 6 × 1 mL of 85% ACN, and the glycan-AEAB conjugate was eluted by 4 × 0.6 mL 100 mM NH_4HCO_3 aqueous solution. Fractions containing glycan-AEAB (monitored by UV Lamp at 365 nm) were collected and lyophilized. The sample was reconstituted in 150 μL of water and loaded on a C18 SPE cartridge (50 mg). The cartridge was washed with 5 × 0.3 mL of water and the glycan-AEAB conjugate was eluted by 4 × 0.6 mL 50% ACN. Fractions containing glycan-AEAB were collected and lyophilized. Glycan-AEAB was quantified by fluorometer (Ex 330 nm, Em 420 nm).

Microarray Printing. MAPA- and AEAB-labeled glycans were reconstituted in phosphate buffer (100 mM sodium phosphate, pH 8.5) at a final concentration of 50 μM and were arrayed onto Nexterion H NHS functionalized slides (Schott AG, Louisville, KY) using a sciFLEXARRAYER S11 from Scienion AG (Berlin, Germany). Each probe was arrayed in four replicates. The average drop volume was within 5% variation of the target volume of 330 pL. After printing, slides were incubated at room temperature in a humidity cabinet set at 80% relative humidity overnight. The slides were then blocked with 50 mM ethanolamine in 100 mM sodium tetraborate buffer (pH 9) for 1 h and washed briefly with 1× phosphate-buffered saline with 0.05% polysorbate 20 (Tween-20). The slides were then briefly rinsed with ultrapure water and dried by centrifugation. The slides were stored desiccated at −20 °C until use.

Microarray Analysis. After rehydration with TSM buffer (20 mM Tris−HCl, 150 mM sodium chloride, 0.2 mM calcium chloride, and 0.2 mM magnesium chloride), the microarray slides were probed with biotinylated lectins and antiblood group antibodies. The bound lectins were detected with SA-Cy5 (0.5 μg/mL). Recombinant human DC-SIGN Fc chimera (10 μg/mL) was detected with Alexa Fluor 488 labeled goat antihuman IgG (H+L, 5 μg/mL). Rat antihuman cutaneous lymphocyte antigen antibody (10 μg/mL) was detected with Alexa Fluor 488 labeled goat antirat IgM (5 μg/mL). Blood Group Lewis a Antibody (7LE, 10 μg/mL) was detected with Alexa Fluor 488 labeled goat antimouse IgG (5 μg/mL). Slides were scanned with a Genepix 4300A microarray scanner from Molecular Devices (Sunnyvale, CA). For cyanine5, Alexa594 and Alexa488, settings of 649 nm (Ex) and 670 nm (Em), 594 nm (Ex) and 645 nm (Em), and 495 nm (Ex) and 519 nm (Em) were used, respectively. Scanned images were quantified using Genepix Pro 7 software.

Reaction System Recycling and Linker Recovery. To a solution of LNnT (20 mg, 28.29 μmol, 1 equiv) in DMSO/HOAc (7/3, 11 mL), F-MAPA (923.39 mg, 2.83 mmol, 100 equiv) and 5MA (1 mg, 5.66 μmol, 0.2 equiv) were added. The mixture was stirred at 65 °C for 4 h. LNnT was completely consumed, and monitored by thin-layer chromatography (TLC) analysis. Then, 10 vols of ethyl acetate (120 mL) was added to the reaction mixture. Ethyl acetate was mixed thoroughly with the reaction mixture and then frozen at −20 °C for 20 min to precipitate LNnT-F-MAPA conjugate. The precipitate was collected by centrifugation and purified by C18 SPE cartridge (2 g) to give 24.7 mg of LNnT-F-MAPA with 86% yield. The supernatant was concentrated to remove ethyl acetate, and 20 mg of LNnT was added to the residue. The mixture was stirred at 65 °C for another 4 h, and then 120 mL ethyl acetate was added to the mixture. Ethyl acetate was mixed thoroughly with the reaction mixture and then frozen at −20 °C for 20 min to precipitate LNnT-F-MAPA conjugate. The precipitate was collected by centrifugation and purified by C18 SPE cartridge (2 g) to give 23.83 mg of LNnT-F-MAPA with 83% yield. Once again, the supernatant was concentrated to remove ethyl acetate, and 20 mg of LNnT was added to the residue. The reaction system was recycled for 5 times with the yield of LNnT-F-MAPA ranging from 86% in the first cycle to 78% in the sixth cycle.

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After 6 cycles, the supernatant was concentrated under reduced pressure to remove acetic acid. Ethyl acetate and 1 M Na_2CO_3 aqueous solution were added to the residue. After phase separation and extraction of the aqueous phase with ethyl acetate, the combined organic phases were dried over Na_2SO_4, concentrated under reduced pressure, and purified by flash chromatography (silica gel column, hexane/ethyl acetate, from 0 to 50%) to provide 738.4 mg of F-MAPA with 80% recovery.

Synthesis of Neoglycoprotein. To a stirred solution of di-*N*-hydroxysuccinimidyl suberate (41.8 mg, 113.6 μmol, 15 equiv) in DMF (1.5 mL), a solution of LNnT-MAPA (6 mg, 7.57 μmol, 1 equiv) and triethylamine (TEA, 114.9 mg, 1.136 mmol, 150 equiv) in 1.5 mL of DMF was added dropwise at 0 °C. The reaction mixture was stirred at room temperature for 3 h. TLC analysis showed complete conversion of LNnT-MAPA. The reaction was then concentrated under high vacuum; the residue was reconstituted in 3 mL of water and extracted with ethyl acetate (3 mL × 3). The aqueous layer was lyophilized to give NHS activated LNnT as a white powder. MALDI TOF-MS: *m*/*z* calcd for $C_{42}H_{70}N_4O_{26}Na$ $[M + Na]^+$, 1046.43; found, 1046.46. The NHS activated LNnT was conjugated to BSA (5 mg/mL) at a molar ratio of 50:1 (25:1, 10:1) in 3 × PBS buffer (pH 7.4). The solution was incubated overnight at room temperature. Then the resulting solution was ultrafiltered and washed with 1 × PBS buffer using Amicon Centrifugal Filter Devices (Ultracel 10 000). The glycoprotein solution was lyophilized to give a white solid. Glycoproteins were analyzed by Bruker ultrafleXtreme MALDI TOF/TOF mass spectrometer.

Derivatization of Egg Yolk *N*-Glycans. As described in blood glycans derivatization, same amount of crude egg yolk *N*-glycans (released by ORNG Method, 1.5 mg) were conjugated with F-MAPA and AEAB separately.

Cleavage of F-MAPA/MAPA Linker. LNnT-F-MAPA (10.15 mg, 10 μmol, 1 equiv) and LNnT-MAPA (7.94 mg, 10 μmol) were dissolved in water (2 mL) separately, and then 4 mg (30 μmol, 3 equiv) and 6.68 mg (50 μmol, 5 equiv) of NCS were added, respectively. The mixtures were stirred at 50 °C for 2 h, and the starting material disappeared completely (monitored by MALDI). Then 2 and 4 μL formic acid were added, respectively, to quench the reaction. The mixtures were lyophilized and purified by P2 biogel filtration to afford 5.8 mg (82% yield) and 5.7 mg (81% yield) LNnT, respectively.

The glycan derivative **20**-F-MAPA (1.57 mg, 1 μmol, 1 equiv) was dissolved in water (100 μL), and then NCS (3 μmol, 3 equiv) was added. The mixture was stirred at 50 °C for 2 h, and the starting material disappeared completely (monitored by MALDI). Then formic acid (3 μmol, 3 equiv) was added to quench the reaction. The mixture was lyophilized and purified by P2 biogel filtration to afford 1.0 mg of **20** with 80% yield.

Release of F-MAPA-labeled egg yolk *N*-glycans was performed as described above. F-MAPA-labeled egg yolk *N*-glycans (0.201 μmol) were dissolved in water (200 μL), and then NCS (0.6 μmol, 3 equiv) was added. The mixture was shaken at 50 °C for 2 h and then washed with chloroform (3 × 0.5 mL). The aqueous phase was detected with fluorometer (Ex 265 nm, Em 315 nM) and the fluorescence disappeared, which suggested that the F-MAPA was completely cleaved. The aqueous phase was concentrated and purified by P2 biogel filtration to afford free reducing egg yolk *N*-glycans.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.bioconjchem.9b00613.

Supporting figures and tables, key resource table, NMR data and spectra, HPLC profiles, mass spectra (PDF)

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Author Contributions

R.DC. and M.W. conceived the project and designed the experiments. M.W. performed the experiments with the assistance of A.Y.M., C.G., N.J., A.M., and L.S. M.W. analyzed the data with the assistance of C.G., T.M., N.J., and J.H.M. R.DC.supervised the project. M.W. and R.DC. wrote the manuscript, which was approved by all authors.

Notes

The authors declare no competing financial interest.

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Dr. Richard D. Cummings (rcummin1@bidmc.harvard.edu).

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